COL761: Data Mining - Assignment 2

Task 1: k-means clustering in high-dimensions — Report

Spandan (2022CS51138) Viraat (2022CS51651) Gaurav (2022CS51144)

March 16, 2026

(a) Reasoning for the selected value of k. Based on the empirical evidence generated by the clustering script (Figure 1), the optimal number of clusters is chosen to be $\boldsymbol{k = 3}$. As k increases from 1 to 3, the k-means objective value (the sum of squared distances or inertia) decreases drastically (from approximately 168,000 to 25,000). Beyond $k = 3$, the rate of decrease flattens out marginally, providing diminishing returns for adding more clusters. This inflection point, known as the "elbow", was detected programmatically by identifying the point on the curve with the maximum perpendicular distance to the line segment connecting the endpoints ($k = 1$) and ($k = 15$).

(b) Structural and statistical assumptions underlying k-means. The k-means algorithm relies on several strong structural and statistical assumptions regarding the data distribution:

- **Spherical Clusters (Isotropy):** It assumes that clusters are convex and structurally spherical in the feature space.
- **Equal Variance:** It assumes all clusters have the same variance/spread.
- **Equal Prior Probabilities:** It implicitly assumes that all clusters have roughly the same density and number of data points.

Statistically, k-means can be viewed as a rigid formulation of a Gaussian Mixture Model (GMM) optimized via Expectation-Maximization, where the covariance matrices of all components are restricted to be strictly spherical and identical ($\sigma^2 I$), and the mixture weights are strictly equal.

(c) Limitations of the model selection procedure. The Elbow Method, while intuitive, has notable limitations:

- **Heuristic Nature:** It lacks a rigorous statistical formulation. In many real-world high-dimensional datasets, the curve is smooth and lacks a sharply defined "elbow", making the selection highly subjective.
- **Focus on Cohesion over Separation:** The objective function only measures intra-cluster variance. It ignores inter-cluster separation, meaning it can sometimes favor configurations where clusters are technically dense but heavily overlapping.

(d) Methodological improvements or alternative validation principles. To overcome the limitations of the Elbow Method, the following principled validation metrics can be utilized:

- **Silhouette Analysis:** Evaluates both intra-cluster cohesion and inter-cluster separation. It returns a score between -1 and 1, providing a robust, standalone metric for selecting k.
- **Gap Statistic:** Compares the total intra-cluster variation for different values of k with their expected values under a null reference distribution of the data (e.g., a uniform distribution over the bounding box of the dataset).
- **Cross-Validation for Clustering:** Evaluates the stability of clusters by splitting the data into folds, clustering them independently, and measuring the consistency of cluster assignments.

(e) Conditions under which k-means is inappropriate and detection. k-means often fails when its underlying assumptions are violated. Well-known failure modes include:

- **Non-convex shapes** (e.g., concentric rings or moons). k-means forces linear boundaries (Voronoi partitions), improperly bisecting these shapes.
- **Anisotropic or varying-density clusters:** k-means will improperly split large, sparse clusters and merge small, dense ones.
- **High-dimensional spaces:** The "curse of dimensionality" causes Euclidean distances between data points to converge, diminishing the algorithm's ability to distinguish intra-cluster vs. inter-cluster distances.

Principled Detection Procedure: To detect these failures in practice, one can employ a secondary generative model like a full-covariance Gaussian Mixture Model (GMM). By comparing the Bayesian Information Criterion (BIC) of the rigid k-means assumptions (GMM with spherical, identical covariance) against a relaxed GMM (independent, unconstrained covariances), a significant improvement in the unconstrained model's BIC serves as a principled indicator that the k-means assumptions are violated. Additionally, uniformly low Silhouette scores across all values of k strongly indicate poorly separated, non-spherical, or highly overlapping clusters.

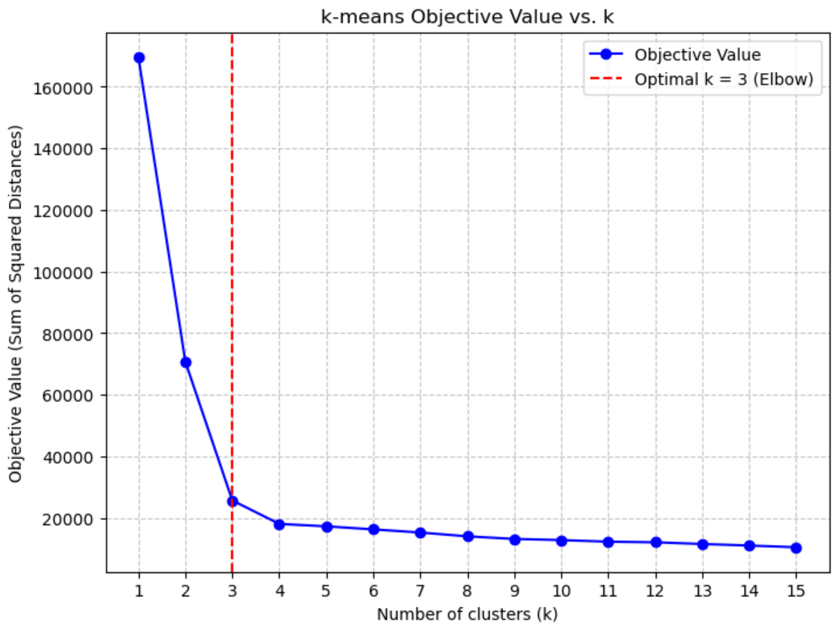

Figure 1: k-means Objective Value vs. k (Empirical Evaluation)

References

[1] Christopher M. Bishop. *Pattern Recognition and Machine Learning.* Springer, 2006.

[2] Robert Tibshirani, Guenther Walther, and Trevor Hastie. "Estimating the number of clusters in a data set via the gap statistic." *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 63.2 (2001): 411-423.